

Master Teorem

Böl ve Yarat

→ Divide and Conquer (Bölme ve Fethet) problemleri için kullanılır.

- Problem a adet n/b boyutta parçaya bölünür. Her adımda k maliyet $f(n)$ ile gösterilir.

$$T(n) = aT(n/b) + f(n)$$

→ Bölme ve Birleştirme
Seydi

↳ a adet parçaya bölünür
↳ a defa çözümlenir.

Ör

$$T(n) = 9T(n/3) + n$$

9 parçaya bölünür
 $n/3$ her birinin boyutu
 $f(n) = n$
 $a = 9$
 $b = 3$

$$\rightarrow n^{\log_b a} = n^{\log_3 9} \rightarrow \Theta(n^2)$$

Kısa Sol Sayfa 11'de
Durum b

Quembs

$T(n)$

$$\Theta(n^{\log_b a})$$

$$\Theta(n^{\log_b a} \log n)$$

$$\Theta(f(n))$$

$$F(n) = O(n^{\log_b a - \epsilon})$$

$$F(n) = \Theta(n^{\log_b a})$$

$$F(n) = \Omega(n^{\log_b a + \epsilon})$$

or

$$aF(n/b) < cF(n)$$

$$T_{\text{anm}} =$$

$$T(n) = aT\left(\frac{n}{b}\right) + F(n)$$

$$a \geq 1$$

$$b > 1$$

$$= 2T\left(\frac{n}{2}\right) + \alpha n$$

Durumlar

$$1-) f(n) = O(n^{\log_b a - \varepsilon})$$

→ Soluğu yoklağırsak

↳ $\varepsilon > 0$ ise

$$\longrightarrow T(n) = \Theta(n^{\log_b a})$$

$$2-) f(n) = \Theta(n^{\log_b a} \log^k n)$$

↳ $k \geq 0$

$$\longrightarrow T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

min değeri için

$$3-) f(n) = \Omega(n^{\log_b a + \varepsilon})$$

↳ $\varepsilon > 0$

$$\longrightarrow T(n) = \Theta(f(n)) \quad \rightarrow c < 1$$

$$a F(n/b) \leq c F(n)$$

①^c

$$T(n) = T\left(\frac{2n}{3}\right) + 1 \quad = ?$$

$$T(n) = 1 \cdot T\left(\frac{n}{3/2}\right) + 1 \rightarrow$$

$$a = 1$$

$$b = 1.5$$

$$f(n) = n^0 = 1$$

$$a \geq 1$$

$$b > 1$$

$$c \leq d, 1$$

Master Theorem:
Gegensatz

$$\log_b a = \log_{3/2} 1 = 0$$

$$n^0 = n^{\log_b a}$$

$$n^0 = n^0 \cdot \log^k n \rightarrow \text{case 2}$$

$k=0$

$$= \Theta\left(\frac{n^{\log_b a}}{n^0} \cdot \log^{k+1} n\right)$$

$$T(n) = \Theta \log n$$

① $T(n) = 16T(n/4) + n$

$a=16$ $b=4$ $F(n)=n^1$ $\log_4 4^2 = 2 //$

$F(n) = ?$

$F(n) = n^{\log_b a} = n^2$

$T(n) = \Theta(n^2)$

← $n = n^{2-\epsilon}$ case 1

↳ algoritmanın maliyeti

② $T(n) = 2T(n/2) + \underline{n \log n}$

$a=2$ $b=2$ $\rightarrow F(n) = n \log n$

$\log_b a = \log_2 2 = 1$

$n \log n = n^1 \cdot \log^k n$
↳ eşitliğin sağlanması

$\log_b a$ case 2 ← için $k=2$ olmalı

$\Theta(n^1 \log^2 n)$

↳ $k+1$

①

$$T(n) = 9T(n/3) + n$$

$$a=9 \quad b=3 \quad f(n)=n$$

$$n^{\log_b a} = n^{\log_3 9} = \underline{n^2} = \underline{f(n)}$$

n^2 'yi bulduk

$f(n)$ ile eşitlik var mı?

Σ ekleme mi?

çıkarma mı?

yada eşit mi?

bu soruları cevaplandır

case'i tespit

ettikten sonra

onu göre işleme

devam edeceğiz

$$n^2 = f(n)$$

$$n^2 \neq n$$

$$n^{2-\epsilon} = n \quad \epsilon=1$$

$\hookrightarrow \Sigma$ ekleme

$\hookrightarrow$ case'i denetle

case'i denetle

$$f(n) = O(n^{\log_b a - \epsilon}) \text{ ise}$$

$$T(n) = \Theta(n^{\log_b a}) \text{ olur}$$

$$T(n) = \Theta(n^{\log_3 9}) = \Theta(n^2)$$

$\hookrightarrow$ Denetle ki bu algoritmanın karmaşıklığı n^2 //

① Merge Sort

$$T(n) = 2T(n/2) + n$$

$$a=2$$

$$b=2$$

$$F(n) = n$$

$$\log_b a = 1$$

$$n \stackrel{?}{=} F(n)$$

$$n = n \quad \checkmark \quad \text{case 2}$$

Case 2 also is:

$$T(n) = \Theta(n^{\log_b a} \log n)$$

$$= \Theta(n^1 \log n)$$

②

$$T(n) = 3T(n/4) + \frac{n \log n}{F(n)}$$

$$a=3 \quad b=4$$

$$n^{\log_b a} = n^{\log_4 3}$$

$$\log_4 3 = ? \quad n^{\log n} \quad \underbrace{\quad}_{F(n)}$$

Case 3

$$\log_4 3 < n \log n$$

→ Bei Kontrollen genau

$$a F(n/b) < c F(n) \quad c \text{ bestimmt } b \text{ und } a$$

$$\underline{3 F(n/4)}$$

$$3 \cdot \frac{1}{4} \log(n/4) < c$$

$$F(n)$$

$$T(n) = \Theta(F(n))$$

$$T(n) = \Theta(n \log n)$$

$\textcircled{or}$ $T(n) = T(n/2) + 2^n$ $\log_2 2 = 2^n$
 $a=1 \quad b=2 \quad F(n)=2^n$ $\textcircled{<}$
Case 3

Kontrol gerekli

$$aF(n/2) < cF(n)$$

$$F \cdot 2^{n/2} < c \cdot 2^{n/2}$$

$$T(n) = F(n) = 2^n$$

$\textcircled{or}$ $T(n) = 64T(n/8) - n^2 \log n$
 $a=64 \quad b=8$ $F(n) = -n^2 \log n$
Master Teoremi: $\log_8 8^2 = ? \quad n^2 \log n$
Uygulanamaz

$\textcircled{or}$ $T(n) = 6T(n/3) + n^2 \log n$
 $a=6 \quad b=3$ $F(n) = n^2 \log n$
 $n^{\log_3 6} = ? \quad F(n)$
 $n^{\log_3 6} = ? \quad \frac{n^2 \log n + \epsilon}{\text{Case 3}}$
 $\rightarrow \Omega$
 $T(n) = F(n)$
 $T(n) = \textcircled{H} (n^2 \log n)$

①

$$3T(n/2) + n^2$$

$$a=3$$

$$b=2$$

$$F(n) = n^2$$

$$n^{\log_2 3 + \epsilon} = ? \quad n^2$$

$\downarrow$
 case 3

$$T(n) = F(n) \quad T(n) = \Theta(n^2)$$

②

$$T(n) = 4T(n/2) + n^2$$

$$a=4$$

$$b=2$$

$$F(n) = n^2$$

$$n^{\log_2 4} = ? \quad n^2$$

$$n^2 = n^2$$

$\leftarrow$
 case 2

$$T(n) = \Theta\left(n^{\log_b a} \cdot \log n\right)$$

$$= \Theta\left(n^2 \cdot \log n\right)$$

③

$$T(n) = \sqrt{2}T(n/2) + \log n$$

$$a=\sqrt{2}$$

$$b=2$$

$$n^{\log_2 \sqrt{2}} = ? \quad \log n$$

$$n^{\frac{1}{2} - \epsilon} = ? \quad \log n$$

$\rightarrow$
 case 1

$$T(n) = \Theta\left(n^{\log_b a}\right)$$

$$= \Theta\left(n^{\log_2 \sqrt{2}}\right) = \Theta\left(n^{\frac{1}{2}}\right) = \Theta\left(\sqrt{n}\right)$$

④

$$T(n) = 0.5T(n/2) + \underline{\underline{1/n}}$$

$a = 1/2$ $b = 2$ $f(n) = 1/n$

$n^{\log_2 1/2} = n^{-1/2}$
 $n^{-1} = ?$ $n^{-1/2}$
 case 2

a > 1
durch

Master Theorem: Ungültig

$$T(n) = \Theta\left(n^{\log_b a} \cdot \log n\right) = \left(\frac{1}{n} \log n\right)$$

$1/n \rightarrow$ zu langsam

⑤

$$T(n) = \underline{\underline{2}} T(n/2) + n^n$$

Master Theorem: Ungültig

$a' \rightarrow "n"$ e abhängig durch
 $\hookrightarrow$ nicht durch

⑥ Merge Sort

$$T(n) = 2T(n/2) + \Theta(n)$$

$a = 2$ $b = 2$ $f(n) = n$

$n^{\log_2 2} = n$
 $n = n$
case 2

$$T(n) = \Theta\left(n^{\log_b a} \cdot \log n\right) = \Theta\left(n \cdot \log n\right)$$

Binary Search Algo.

Q

$$T(n) = T(n/2) + O(1)$$

$$a=1 \\ b=2$$

$$A_n = 1$$

$$n^{\log_2 1} = ?$$

$$1^0 = ?$$

$$1 = 1$$

case 2

$$T(n) = O(\log_b a \cdot \log n)$$

$$= O(1 \cdot \log n)$$

Thisa Yoi

Q

$$T(n) = aT(n/b) + cn^d \quad (\text{for } n > 1)$$

$$a \geq 1 \quad b \geq 2 \quad c > 0 \quad d \geq 0$$

sorter

$$\text{if } a < b^d \text{ ise } T(n) = O(n^d)$$

$$\text{if } a = b^d \text{ ise } T(n) = O(n^d \cdot \log n)$$

$$\text{if } a > b^d \text{ ise } T(n) = O(n^{\log_b a})$$

~~Q5~~ $2T(n/4) + 1 = n^0$

Normal

$a=2 \quad b=4 \quad F(n)=1$

$n^{\log_4 2} = ? \quad \downarrow \quad n^{1/2} = ? \quad \uparrow$

Case 1

$T(n) = \Theta(n^{\log_4 2}) = n^{\log_4 2}$
 $n^{\log_4 4^{1/2}} = n^{1/2}$

$\downarrow$ $\sqrt{2} \quad \sqrt{4}$
 $a \geq 1 \quad b \geq 2$
 $c > 0 \quad d \geq 0$
 $\hookrightarrow 1 \quad \hookrightarrow 0$

$\rightarrow 2 ? 4^0$

$2 > 1$

$T(n) = \Theta(n^{\log_4 2})$
 $n^{1/2}$

~~Q5~~ $T(n) = 2T(n/4) + \sqrt{n}$

Normal $a=2 \quad b=4 \quad c=1 \quad d=1/2$
 $F(n) = n^{1/2}$

$n^{\log_4 a} = ? \quad F(n)$

$n^{\log_4 4^{1/2}}$

$n^{1/2} = ? \quad n^{1/2}$

Case 2

$T(n) = \Theta(\log_b a \cdot \log n)$
 $= \Theta(\sqrt{n} \log n)$

Case

$2 ? 4^{1/2}$

$2 ? \sqrt{4}$

$2 = 2 \rightarrow \text{equal}$

$T(n) = \Theta(n^d \cdot \log n)$

$= \Theta(n^{1/2} \cdot \log n)$

